COMP3670 MACHINE LEARNING

LEARNING / STATISTICS / MODELS / CLUSTERING / ORTHONORMAL GEOMETRY

Learning from data

Deduction: theory/premises → prediction; true premises and valid logic imply the conclusion. **Induction:** observations → a general rule; finite evidence cannot prove a universal claim. One counterexample can falsify it. **Supervised:** labelled (x_i, y_i); classification (discrete y), regression (numerical y). **Unsupervised:** only x_i; discover structure. **Reinforcement:** state → action → reward; learn a policy maximizing cumulative reward, often delayed.

PLA ($y_i \in \{-1, +1\}$). Initialize $w = b = 0$. Find a misclassified point, update, repeat until no errors:

$$w \leftarrow w + y_i x_i, \qquad b \leftarrow b + y_i.$$

L1 predicts $+1$ if $w^T x + b > 0$, otherwise -1. Strictly linearly separable data ⇒ finitely many updates; otherwise no termination guarantee. Zero training error does not imply generalization.

Variance & numerical integration

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2.$$

For $I = \int_a^b f(x)dx$, let $h = (b-a)/n$, $x_i = a + ih$, $f_i = f(x_i)$.

Composite trapezoidal rule

$$T_n = \frac{h}{2}\left[f_0 + 2\sum_{i=1}^{n-1} f_i + f_n\right].$$

If $\max_{[a,b]} |f''| \le M_2$:

$$|I - T_n| \le \frac{M_2(b-a)^3}{12n^2}.$$

Composite Simpson rule (n even)

$$S_n = \frac{h}{3}\left[f_0 + 4\sum_{\substack{1\le i<n \\ i \text{ odd}}} f_i + 2\sum_{\substack{2\le i<n \\ i \text{ even}}} f_i + f_n\right].$$

If $\max |f^{(4)}| \le M_4$, $|I - S_n| \le M_4(b-a)^5/(180n^4)$. Doubling n: error bounds divide by 4 (trapezoid), 16 (Simpson).

Generalization & complexity

$D = \{(X_i, Y_i)\}_{i=1}^N$ i.i.d.; $L(h) = E_{out}(h)$, $\widehat{L}(h) = E_{in}(h)$; ERM minimizes $\widehat{L}$. Bounds below use binary 0–1 loss.

Hoeffding, fixed h before viewing D:

$$\Pr\{|L(h) - \widehat{L}(h)| > \epsilon\} \le 2e^{-2N\epsilon^2}.$$

For finite $|\mathcal{H}| = M$, union bound gives

$$\Pr\{\sup_h |L(h) - \widehat{L}(h)| > \epsilon\} \le 2Me^{-2N\epsilon^2}.$$

Thus simultaneously $L(h) \le \widehat{L}(h) + \sqrt{\log(2M/\delta)/(2N)}$, with probability $\ge 1 - \delta$. Searching over data-selected h requires uniform control.

Shatter: realize all 2^N labellings of some N points. $S(\mathcal{H}, N)$: maximum number of labellings; $v = d_{VC}$: largest shatterable size. One-direction threshold: 1; intervals: 2; affine hyperplanes in $\mathbb{R}^d$: $d + 1$.

Sauer: finite v, $N > v \Rightarrow S(\mathcal{H}, N) \le (N+1)^v$. Course VC bound:

$$\Pr\{\sup_h |L - \widehat{L}| > \epsilon\} \le 8(N+1)^v e^{-N\epsilon^2/32}.$$

Penalty $\sqrt{\frac{32}{N}[v\log(N+1) + \log(8/\delta)]}$.

Agnostic PAC: $\Pr\{L(\hat{h}) \le L(h^*) + \epsilon\} \ge 1 - \delta$ for all P, once $N \ge N_{\mathcal{H}}(\epsilon, \delta)$; $h^* = \arg\min_{h\in\mathcal{H}} L(h)$. Binary distribution-free PAC ⇔ $v < \infty$ (standard measurability assumptions). Sufficient course-bound order:

$$N_{\mathcal{H}} = O\left(\frac{v\log(1/\epsilon) + \log(1/\delta)}{\epsilon^2}\right).$$

Approximation vs estimation:

$$\underbrace{L(\hat{h}) - L(f^*)}_{\text{excess over Bayes}} = \underbrace{L(h^*) - L(f^*)}_{\Delta_{app}} + \underbrace{L(\hat{h}) - L(h^*)}_{\Delta_{est}}.$$

Restricting $\mathcal{H}$ can increase approximation error but reduce estimation difficulty. This is related to the bias/variance trade-off.

K-fold CV: fit on $K-1$ folds, score the held-out fold; rotate, then average:

$$E_{CV}(M) = K^{-1}\sum_{j=1}^{K} E_{val}^{(j)}(M).$$

Choose hyperparameters by CV, refit on development data; test once on untouched data. Fit preprocessing inside each fold. Fold scores are dependent; CV winner's score can be optimistic.

Parametric estimation & logistic models

Parametric: fixed finite-dimensional family.
Nonparametric: complexity may grow with data; locality/smoothing replaces a fixed shape assumption.
Plug-in: estimate η, then apply a decision rule.
Likelihood / MLE (i.i.d. observations):

$$\mathcal{L}(\theta) = \prod_{i=1}^N p(z_i \mid \theta),$$

$$\hat{\theta} = \arg\max_\theta \sum_i \log p(z_i \mid \theta).$$

Data fixed, parameter varies; likelihood is not a probability distribution over θ. Bernoulli: $\hat{p} = k/N$. Gaussian MLE:

$$\hat{\mu} = N^{-1}\sum_i z_i, \qquad \hat{\sigma}^2 = N^{-1}\sum_i (z_i - \hat{\mu})^2.$$

Consistency/efficiency need correct specification and regularity; a wrong family can fit well yet remain wrong.
Logistic regression, $y \in \{0, 1\}$:

$$g = w^T x + b, \quad p = \sigma(g) = \frac{1}{1 + e^{-g}},$$

$$\log\frac{p}{1-p} = g.$$

Odds $= e^g$; increasing x_j by one multiplies odds by e^{w_j}, holding others fixed. Threshold $p > 1/2$ iff $g > 0$. Costs can change the threshold.

$$\ell_{CE}(y, p) = -y\log p - (1-y)\log(1-p).$$

Conditional likelihood $\prod_i p_i^{y_i}(1-p_i)^{1-y_i}$; MLE = minimize mean cross-entropy, not directly 0–1 error. Confident wrong probabilities incur large loss.
Logistic gradient (augmented x includes bias):

$$\sigma'(z) = \sigma(z)(1 - \sigma(z)),$$

$$\nabla_w \ell_i = (\sigma(w^T x_i) - y_i)x_i,$$

$$\nabla J = \frac{1}{N}\sum_i (p_i - y_i)x_i, \qquad \nabla^2 J = \frac{1}{N}X^T D X,$$

$D = \mathrm{diag}(p_i(1-p_i)) \succeq 0$. Convex, usually no closed form; unregularized separable data may have no finite MLE.

Nonparametric classification

Histogram density, bin $A(x)$, width l in 1D:

$$\hat{f}(x) = \frac{\#\{i : X_i \in A(x)\}}{Nl}.$$

In d dimensions, divide by bin volume instead of l. **Kernel density estimate**, normalized κ:

$$\hat{f}(x) = \frac{1}{N\sigma^d}\sum_i \kappa\left(\frac{x - X_i}{\sigma}\right).$$

Small bins/bandwidth: high variance; large: more smoothing/bias. Scaling and irrelevant features affect distances; high dimension causes sparse neighborhoods.
Unified local vote ($W_i \ge 0$, $\sum_i W_i = 1$):

$$\hat{\eta}(x) = \sum_i W_i(x)\mathbf{1}\{Y_i = 1\},$$

$$\hat{h}(x) = \mathbf{1}\{\hat{\eta}(x) > 1/2\}.$$

Histogram: $W_i = \mathbf{1}\{X_i \in A(x)\}/N(x)$; handle empty cells. **kNN:** $W_i = 1/k$ for the k nearest points, else 0. Small k: flexible/noisy. Odd k prevents binary vote ties, not distance ties.

$$\text{Kernel:} \quad W_i(x) = \frac{\kappa((x - X_i)/\sigma)}{\sum_j \kappa((x - X_j)/\sigma)}.$$

Handle a zero denominator for compact-support kernels. Gaussian $e^{-\|u\|^2}$; Cauchy $(1 + \|u\|^{d+1})^{-1}$; uniform $\mathbf{1}_{\|u\|\le 1}$; triangular $(1 - \|u\|)_+$; Epanechnikov $(1 - \|u\|^2)_+$. Common constants cancel in normalized weights. Select k, l, σ by validation/CV.

Trees, bagging & random forests

CART: recursively split $x_j \le a$; leaves form axis-aligned boxes. Predict majority class in each leaf. Each path is an AND; same-class leaves are OR'ed.
Let p be class-1 fraction in node R:

$$I_G = 2p(1-p), \quad I_{err} = \min(p, 1-p),$$

$$I_H = -p\log p - (1-p)\log(1-p), \quad 0\log 0 = 0.$$

Impurity 0 at pure nodes, maximal at $p = 1/2$.

$$\Delta I = I(R) - \frac{N_L}{N_R}I(R_L) - \frac{N_U}{N_R}I(R_U).$$

Here N_R is parent size, N_L, N_U are child sizes. Choose the largest admissible decrease; repeat. A local greedy split is not global ERM over all trees. Stop by purity, depth, minimum leaf size; prune/validate. Deep single trees have high variance. Monotone feature rescaling preserves threshold order.

K-means

For nonempty clusters $C_1, \ldots, C_k$:

$$J = \sum_{r=1}^k \sum_{i\in C_r} \|x_i - c_r\|^2, \qquad c_r = \frac{1}{|C_r|}\sum_{i\in C_r} x_i.$$

Lloyd steps: initialize centers/partition; assign each point to its nearest center; recompute means; repeat until assignments stabilize. Use multiple starts; different initializations can reach different local optima.

$$\sum_i \|x_i - c\|^2 = \sum_i \|x_i - \bar{x}\|^2 + n\|c - \bar{x}\|^2.$$

Thus the mean uniquely minimizes a nonempty cluster's SSE. Assignment and mean updates never increase J. With no empty clusters and ties kept in the current cluster, any assignment change strictly decreases J; finitely many partitions ⇒ finite termination, not global optimality.
Empty cluster: its mean is undefined; repair/reseed (e.g. a distant point), or keep the previous center under a chosen policy. The stated termination proof assumes nonempty clusters, so do not silently divide by zero. Feature scaling changes the geometry.

Hierarchical clustering

Agglomerative: start with singleton clusters; repeatedly merge the closest pair under a chosen linkage; record merge height; continue to one cluster.

$$d_{single}(A, B) = \min_{x\in A, y\in B} d(x, y),$$

$$d_{complete}(A, B) = \max_{x\in A, y\in B} d(x, y),$$

$$d_{average}(A, B) = \frac{1}{|A||B|}\sum_{x\in A}\sum_{y\in B} d(x, y).$$

Single can chain via bridges; complete favors compact clusters; average uses all cross-pairs. A dendrogram's vertical height is merge dissimilarity. A horizontal cut gives clusters as connected branches below it; horizontal spacing has no distance meaning. Linkage choice can change the partition at the same cut height.

Orthonormal geometry

Cauchy–Schwarz:

$$|x^T y| \le \|x\|\|y\|.$$

Equality iff x, y linearly dependent (including a zero vector). Proof: $0 \le \|x - ay\|^2$, choose $a = x^T y/\|y\|^2$ for $y \ne 0$.
Orthonormal list: $q_i^T q_j = \delta_{ij}$, hence linearly independent. If it spans S, it is an orthonormal basis. With $Q = [q_1 \ \cdots \ q_k]$:

$$Q^T Q = I_k, \qquad [x]_{\mathcal{Q}} = Q^T x \quad (x \in S).$$

The coefficients follow by taking dot products. Rectangular Q generally has $QQ^T \ne I_d$.
Orthogonal matrix is square with $Q^T Q = I$:

$$Q^{-1} = Q^T, \quad (Qx)^T(Qy) = x^T y, \quad \|Qx\| = \|x\|.$$

It preserves lengths, distances and nonzero-vector angles. Do not confuse orthogonal columns with a square orthogonal matrix.
Gram–Schmidt, independent $v_1, \ldots, v_k$:

$$u_1 = v_1, \quad q_1 = u_1/\|u_1\|,$$

$$u_j = v_j - \sum_{i<j}(q_i^T v_j)q_i, \qquad q_j = u_j/\|u_j\|.$$

Subtract *all* previous components, then normalize. Prefix spans are unchanged. Zero residual means dependence; skip it rather than normalize zero. Check $Q^T Q = I$ afterwards.

Spectral theorem & PSD

For real symmetric $A = A^T \in \mathbb{R}^{d\times d}$, there is a real orthonormal eigenbasis:

$$Aq_j = \lambda_j q_j, \quad Q^T Q = I,$$

$$A = Q\Lambda Q^T = \sum_{j=1}^d \lambda_j q_j q_j^T.$$

Distinct-eigenvalue eigenvectors are orthogonal; within a repeated eigenspace choose an orthonormal basis. Q^T: obtain coordinates; Λ: scale; Q: reconstruct.

$$x = \sum_j a_j q_j \quad \Rightarrow \quad x^T A x = \sum_j \lambda_j a_j^2.$$

Positive semidefinite: $A \succeq 0$ iff $x^T Ax \ge 0$ for all x, iff every $\lambda_j \ge 0$. **Positive definite:** strict positivity for every nonzero x, iff every $\lambda_j > 0$.
PSD can be singular; PD is invertible. For any real B, $B^T B \succeq 0$ since $x^T B^T Bx = \|Bx\|^2$. Negative eigenvalues rule out PSD. A general real non-symmetric matrix need not have an orthonormal eigenbasis.

SVM: hard, soft & nonlinear

Labels $y_i \in \{-1, +1\}$; $f_i = w^T x_i + b$. SVM convention: predict $+1$ for $f \geq 0$, else -1. With $w \neq 0$:

$$m_i = y_i f_i, \quad \gamma_i = \frac{m_i}{\|w\|}, \quad d(x, H) = \frac{|w^T x + b|}{\|w\|}.$$

Positive rescaling preserves boundary/margins, not raw scores.

Hard margin (both classes, strictly separable):

$$\min_{w,b} \tfrac{1}{2}\|w\|^2 \quad \text{s.t. } m_i \geq 1 \ (\forall i).$$

At optimum $\min_i m_i = 1$, training margin $\gamma = 1/\|w\|$, full band width $2/\|w\|$. **Support vectors:** points with $m_i = 1$ touch the margin boundaries.

Soft margin, $C > 0$:

$$\min_{w,b,\xi} \quad \tfrac{1}{2}\|w\|^2 + C\sum_i \xi_i,$$

$$\text{s.t.} \quad m_i \geq 1 - \xi_i, \quad \xi_i \geq 0.$$

For fixed w, b, $\xi_i^* = (1 - m_i)_+ = \max(0, 1 - m_i)$; equivalent objective:

$$J(w, b) = \tfrac{1}{2}\|w\|^2 + C\sum_i (1 - m_i)_+.$$

$\xi = 0$: satisfies margin; $0 < \xi < 1$: correct side inside band; $\xi = 1$: decision boundary; $\xi > 1$: wrong side. Correct labels may still incur hinge loss. In soft SVM, inside-margin/violating points can influence the fit too. Large C: penalize violations more; small C: prefer small norms. Feature units change distances and regularization geometry; tune scaling and C together. Soft margin allows $w = 0$, a constant score.

QP: quadratic convex objective plus linear constraints; every local minimum is global.

Subgradient step. Compute $I = \{i : m_i < 1\}$ from the same old parameters, then update both:

$$w^+ = (1 - \eta)w + \eta C \sum_{i \in I} y_i x_i,$$

$$b^+ = b + \eta C \sum_{i \in I} y_i.$$

Shrink weights and correct violations. At $m = 1$, zero hinge subgradient is valid. A step may increase J; arbitrary constant steps need not converge.

Nonlinear SVM: choose a fixed feature map ϕ; replace x_i by $\phi(x_i)$ in the score, hinge objective and updates. Still convex in (w, b); curved boundary in original space. E.g. $\phi(x) = (x, x^2)$, $4x - x^2 - 3 > 0 \iff 1 < x < 3$. A radial feature $x_1^2 + x_2^2$ represents circular boundaries.

SVD & latent-factor recommendation

$A \in \mathbb{R}^{m \times d}$, rank r; compact SVD:

$$A = U_r \Sigma_r V_r^T = \sum_{j=1}^{r} \sigma_j u_j v_j^T,$$

$U_r \in \mathbb{R}^{m \times r}, V_r \in \mathbb{R}^{d \times r}$ have orthonormal columns; $\sigma_1 \geq \cdots \geq \sigma_r > 0$.

$$Av_j = \sigma_j u_j, \quad A^T u_j = \sigma_j v_j, \quad A^T A v_j = \sigma_j^2 v_j.$$

Truncation: $A_k = U_k \Sigma_k V_k^T$ is best among rank-$\leq k$ matrices in Frobenius error:

$$\|A - A_k\|_F^2 = \sum_{j>k} \sigma_j^2, \quad \|A\|_F^2 = \sum_j \sigma_j^2.$$

Ratings: $R \in \mathbb{R}^{n \times m}$, $P \in \mathbb{R}^{n \times k}, Q \in \mathbb{R}^{m \times k}$; $\hat{r}_{ij} = p_i^T q_j$, $\hat{R} = PQ^T$. Fully observed: $P = U_k \Sigma_k, Q = V_k$. Rows are individual embeddings; columns are patterns.

Missing ratings: unknown $\neq 0$. Fit only observed training pairs Ω:

$$L = \sum_{(i,j) \in \Omega} (r_{ij} - p_i^T q_j)^2 + \lambda(\|P\|_F^2 + \|Q\|_F^2).$$

ALS ($\lambda > 0$): small random initialization; alternate exact ridge solves. Let Q_i contain item-factor rows rated by user i, r_i their ratings; P_j contains users rating item j, with ratings r_j:

$$p_i = (Q_i^T Q_i + \lambda I)^{-1} Q_i^T r_i,$$

$$q_j = (P_j^T P_j + \lambda I)^{-1} P_j^T r_j.$$

Solve systems, do not explicitly invert. Exact blocks do not increase L; joint problem nonconvex, no global guarantee. Validate k, λ; untouched test ratings assess prediction. No training records: cold start.

$$\text{RMSE} = \sqrt{\frac{1}{|T|} \sum_{(i,j) \in T} (r_{ij} - \hat{r}_{ij})^2}.$$

Second-order tests & convexity

1D stationary point: $f'(x^*) = 0$ (interior, differentiable) is necessary, not sufficient. If $f''(x^*) > 0$: strict local min; < 0: strict local max; $= 0$: inconclusive. Check boundaries and non-differentiable points too.

Multivariable: $\nabla f(x^*) = 0$; Hessian $H_{ij} = \partial_i \partial_j f$ (symmetric for C^2 functions).

$$f(x + v) \approx f(x) + \nabla f(x)^T v + \tfrac{1}{2} v^T H(x) v.$$

At a stationary point: $H \succ 0$ implies strict local min; $H \prec 0$ implies strict local max; indefinite H implies saddle. Semidefinite/singular H: test inconclusive. Local min requires $H \succeq 0$ for C^2 functions.

2D test at a stationary point, $D = f_{xx} f_{yy} - f_{xy}^2$: $D > 0, f_{xx} > 0$ gives a min; $D > 0, f_{xx} < 0$ a max; $D < 0$ a saddle; $D = 0$ is inconclusive.

Convexity on a convex domain:

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y), \quad t \in [0, 1].$$

For differentiable f, equivalent to $f(y) \geq f(x) + \nabla f(x)^T (y - x)$. For C^2 f, equivalent to $H(x) \succeq 0$ everywhere. Every local min is global; $\nabla f = 0$ is sufficient for global min. Strict convexity gives at most one minimizer, not its existence. $H \succ 0$ everywhere is sufficient for strict convexity, not necessary.

Normal equations & ridge

Rows of design $X \in \mathbb{R}^{N \times p}$ include any intercept/basis features; $y \in \mathbb{R}^N$.

$$RSS = \|X\theta - y\|^2, \quad \nabla RSS = 2X^T(X\theta - y),$$

$$\nabla^2 RSS = 2X^T X \succeq 0, \qquad X^T X \hat{\theta} = X^T y.$$

If X has full column rank:

$$\hat{\theta} = (X^T X)^{-1} X^T y.$$

Otherwise minimizers need not be unique; fitted training predictions are unique. Numerically solve least squares, not an explicit inverse.

Ridge with $\lambda > 0$:

$$J = RSS + \lambda\|\theta\|^2, \quad \hat{\theta} = (X^T X + \lambda I)^{-1} X^T y,$$

$$\nabla J = 2X^T(X\theta - y) + 2\lambda\theta.$$

$X^T X + \lambda I \succ 0$: unique solution and improved conditioning. Formula penalizes all entries; to leave intercept unpenalized replace I by $\text{diag}(0, 1, \ldots, 1)$ (uniqueness then needs corresponding conditions).

Parametric vs local regression. Linear parameter models use $h_\theta(x) = \sum_j \theta_j \phi_j(x)$; features may be nonlinear in x. For squared loss, $f^*(x) = \mathbb{E}[Y \mid X = x]$. **Kernel / kNN regression:** $\hat{h}(x) = \sum_i W_i(x) Y_i$, using normalized kernel weights or $1/k$ for nearest neighbors.

Regression trees: leaf mean, splits minimizing child RSS; forests average predictions.

GD, learning rates & convergence

$$\theta_{t+1} = \theta_t - \alpha \nabla J(\theta_t).$$

Full GD uses all N examples per step. For $J = RSS/N$:

$$\theta^+ = \theta - \alpha \frac{2}{N} X^T(X\theta - y).$$

Large α may oscillate/diverge; tiny α is slow. Stop by small gradient, small objective/parameter changes, or iteration budget; validation can guide early stopping.

Convex K-smooth objective, minimizer θ^* exists and $0 < \alpha \leq 1/K$:

$$J(\theta_t) - J(\theta^*) \leq \frac{\|\theta_0 - \theta^*\|^2}{2\alpha t}.$$

K bounds Hessian eigenvalues / gradient Lipschitz constant. For strongly convex J with curvature $\mu > 0$, suitable GD has a linear rate. Nonconvex descent need not find a global minimum.

SGD / mini-batches, $J = N^{-1} \sum_i \ell_i$:

$$g_t = \frac{1}{b} \sum_{i \in B_t} \nabla \ell_i(\theta_t), \quad \theta_{t+1} = \theta_t - \alpha_t g_t.$$

Uniform sampling gives $\mathbb{E}[g_t \mid \theta_t] = \nabla J(\theta_t)$. One epoch $\approx N/b$ updates. Smaller b: cheaper, noisier steps; larger b: less gradient variance, more cost. Apply regularizer gradients separately, without dropping their scaling.

A fixed learning rate often leaves a noise floor. Under standard stochastic-approximation assumptions, a sufficient diminishing-step condition is

$$\sum_t \alpha_t = \infty, \qquad \sum_t \alpha_t^2 < \infty.$$

E.g. $\alpha_t = a/(b + t)^\gamma$, $1/2 < \gamma \leq 1$. These step rules alone do not guarantee convergence without assumptions on the objective and gradient noise.

Ensemble learning

Bagging: draw B bootstrap samples of size N with replacement; fit a tree per sample; majority vote (classification), average (regression). Variance reduction is greater when members are less correlated.

Random forest: bootstrap plus a random feature subset *at every split*; often deep trees. Feature randomness decorrelates trees. Forests are more stable, less interpretable; no universal immunity to overfitting.

Neural networks & backpropagation

Forward pass: $a^0 = x$; hidden layers

$$z^\ell = W^\ell a^{\ell-1} + b^\ell, \qquad a^\ell = \phi(z^\ell).$$

Without nonlinear activations, stacked affine layers collapse to one affine map. Training is generally nonconvex; expressiveness does not guarantee generalization.

Activations: ReLU $(z)_+ = \max(0, z)$; $\sigma(z) = 1/(1 + e^{-z})$; $\tanh z$; $\arctan z$; step $\mathbf{1}\{z \geq 0\}$.

$$\phi'_{ReLU} = \mathbf{1}\{z > 0\} \ (z \neq 0), \quad \sigma' = \sigma(1 - \sigma),$$

$$(\tanh z)' = 1 - \tanh^2 z, \quad (\arctan z)' = (1 + z^2)^{-1}.$$

ReLU at zero: choose a subgradient (often 0); step has no useful ordinary gradient at its jump.

Output: binary sigmoid + binary cross-entropy; multiclass softmax + cross-entropy:

$$p_c = \frac{e^{z_c}}{\sum_j e^{z_j}}, \quad \hat{y} = \arg\max_c p_c,$$

$$\ell = -\sum_c y_c \log p_c.$$

Use one-hot y; subtract $\max_j z_j$ before exponentiating for numerical stability. For either matched sigmoid/BCE or softmax/CE, output error is $\delta^L = p - y$.

Backprop (linear output logits):

$$\delta^\ell = ((W^{\ell+1})^T \delta^{\ell+1}) \odot \phi'(z^\ell),$$

$$\frac{\partial \ell}{\partial W^\ell} = \delta^\ell (a^{\ell-1})^T, \qquad \frac{\partial \ell}{\partial b^\ell} = \delta^\ell.$$

Average sample gradients for mean batch loss. Chain rule transmits error backwards; keep tensor shapes consistent. Saturating activations can yield small gradients.

Initialization: small random zero-mean weights break neuron symmetry; biases may be zero. Weight variances (not standard deviations), fan-in n_{in}, fan-out n_{out}:

$$\text{LeCun } \frac{1}{n_{in}}, \quad \text{Xavier } \frac{2}{n_{in} + n_{out}}, \quad \text{He } \frac{2}{n_{in}}.$$

He suits ReLU; Xavier balances layer scales. All-zero hidden weights keep units symmetric.

Optimizer update equations

Let $g_t = \nabla J(w_t)$ or a mini-batch gradient; all vector squares, roots, products/divisions below are coordinatewise. Initialize historical states to zero. $0 < \beta, \beta_1, \beta_2 < 1$; $\epsilon > 0$ stabilizes denominators.

Momentum / heavy ball (velocity convention):

$$v_t = \beta v_{t-1} - \alpha_t g_t, \qquad w_{t+1} = w_t + v_t.$$

Accumulates descent directions; can reduce oscillation.

Nesterov (gradient at look-ahead position):

$$v_t = \beta v_{t-1} - \alpha_t \nabla J(w_t + \beta v_{t-1}),$$

$$w_{t+1} = w_t + v_t.$$

Evaluate the gradient at the look-ahead position.

Adagrad (permanent squared-gradient sum):

$$s_t = s_{t-1} + g_t \odot g_t, \qquad w_{t+1} = w_t - \frac{\alpha_t g_t}{\sqrt{s_t + \epsilon}}.$$

Effective coordinatewise learning rates shrink; useful for sparse coordinates, but may become too small.

RMSProp (EMA instead of permanent sum):

$$s_t = \beta s_{t-1} + (1 - \beta) g_t \odot g_t,$$

$$w_{t+1} = w_t - \frac{\alpha_t g_t}{\sqrt{s_t + \epsilon}}.$$

Forgets old squared gradients. Course convention puts ϵ inside the root in these two methods.

Adam (course's *negative-gradient* moment):

$$m_t = \beta_1 m_{t-1} - (1 - \beta_1) g_t,$$

$$s_t = \beta_2 s_{t-1} + (1 - \beta_2) g_t \odot g_t,$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \qquad \hat{s}_t = \frac{s_t}{1 - \beta_2^t}, \qquad t \geq 1,$$

$$w_{t+1} = w_t + \frac{\alpha_t \hat{m}_t}{\sqrt{\hat{s}_t + \epsilon_{root}} + \epsilon}.$$

Use $m_0 = s_0 = 0$; first gradient is $t = 1$ for bias correction. Common course values: $\beta_1 = .9$, $\beta_2 = .999$, $\epsilon = 10^{-8}$, $\epsilon_{root} = 0$.

Sign check: negative gradient is already stored in m_t, so Adam's final update is *plus*. With a positive-gradient first moment, use a minus update instead. Do not mix conventions. Bias correction compensates zero-initialized EMAs, not statistical model bias.